

C Programming Laboratory Assignments

Complete Collection of 10 Labs

Lab 1: Basics (Input/Output, Operators, Conversion)

1. Write a program to accept marks of 5 subjects, find the sum and calculate percentage.
2. Write a program to calculate Simple Interest and Compound Interest.
3. Write a program to calculate the area and circumference of a circle for a given radius.
4. Write a program to convert temperature between Celsius and Fahrenheit.
5. Write a program to convert distance from kilometers to miles and miles to kilometers.
6. Write a program to convert days into years, months and remaining days (assume 365 days = 1 year, 30 days = 1 month).
7. Write a program to calculate HRA, DA, and Gross Salary for a given basic salary.
8. Write a program to find the perimeter of a rectangle and the area of a square.
9. Write a program to perform all arithmetic operations (+, -, ×, ÷, %) on two numbers.
10. Write a program to check whether a given character is a vowel or consonant.

Lab 2: Mathematical Foundation of Problem Solving

1. Write a C program to check whether a number is an Armstrong number.
2. Write a C program to check whether a number is a palindrome.
3. Write a C program to find the sum of digits of a number.
4. Write a C program to check whether a number is a Strong number.
5. Write a C program to check whether a number is a Perfect number.
6. Write a C program to check whether three numbers form a Pythagorean triplet.
7. Write a C program to count the frequency of each digit (0–9) in a number.
8. Write a C program to check whether a number is an Automorphic number.
9. Write a C program to check whether a number is a Harshad (Niven) number.
10. Write a C program to check whether a number is a Cyclic number.
11. Write a C program to find the sum of the series $1 + 11 + 111 + \dots$ up to N terms.
12. Write a C program to find the missing number in a sequence from 1 to N.

13. Write a C program to check whether a number is a Happy number.

Lab 3: Conditions (if-else, relational operators)

1. Write a program to swap values of two variables using a third variable.
2. Write a program to check whether two numbers are equal or not.
3. Write a program to find the greatest among three numbers.
4. Write a program to check whether a given number is even or odd.
5. Write a program to find the roots of a quadratic equation.
6. Write a program to check whether a given character is an alphabet, digit, or special symbol.
7. Write a program to check whether a number is positive, negative or zero.
8. Write a program to check whether three sides can form a valid triangle and display its type.
9. Write a program to calculate electricity bill based on units consumed (using slab rates).
10. Write a program to create a simple calculator using if-else ladder.

Lab 4: Decisions & Switch

1. Write a program to check whether a given year is a leap year or not.
2. Write a program to accept marks of 5 subjects, calculate percentage and display grade (A/B/C/D).
3. Write a program to create a calculator using switch-case.
4. Write a program to display the weekday name for a given day number (1–7).
5. Write a program to display the number of days in a given month (1–12).
6. Write a program to check whether a character is vowel or consonant using switch.
7. Write a program to check eligibility for voting ($\text{age} \geq 18$).
8. Write a program to categorize a person based on age (Child/Teen/Adult/Senior).
9. Write a program to create a menu-driven program to calculate area of geometric shapes.
10. Write a program to create a menu-driven program for temperature conversion ($C \leftrightarrow F$).

Lab 5: Loops (for, while, do-while)

1. Write a program to find the sum of natural numbers up to N.
2. Write a program to find the factorial of a given number.
3. Write a program to calculate the sum of even and odd numbers up to N.
4. Write a program to print the multiplication table of a given number.

5. Write a program to print the first N natural numbers.
6. Write a program to reverse the digits of a number using loop.
7. Write a program to count the number of digits in a number.
8. Write a program to check whether a number is a palindrome or not.
9. Write a program to print all factors of a given number.
10. Write a program to calculate power (x^y) without using the pow() function.

Lab 6: Series & Recursion

1. Write a program to generate Fibonacci series up to N terms (loop).
2. Write a program to check whether a number is prime or not.
3. Write a program to find the sum of digits of a number.
4. Write a program to check whether a number is an Armstrong number.
5. Write a program to check whether a number is a Strong number.
6. Write a program to check whether a number is a Perfect number.
7. Write a program to print all prime numbers in the range 1–N.
8. Write a program to generate Fibonacci series using recursion.
9. Write a program to find factorial of a number using recursion.
10. Write a program to find GCD and LCM of two numbers.
11. Write a program to reverse a number using recursion.

Lab 7: Patterns

1. Write a program to print all Armstrong numbers between 1 and 100.
2. Write a program to convert a binary number to decimal and decimal to binary.
3. Write a program to print Pascal's Triangle.
4. Write a program to print Floyd's Triangle.
5. Write a program to print a pyramid of numbers.
6. Write a program to print a pyramid of stars.
7. Write a program to print a diamond pattern.
8. Write a program to print a hollow square pattern.
9. Write a program to print an inverted half pyramid.

Lab 8: Arrays (1D, Searching, Sorting)

1. Write a program to accept elements in an array and find their sum.
2. Write a program to add two arrays and store result in a third array.
3. Write a program to find the minimum and maximum element of an array.
4. Write a program to search an element in an array using Linear Search.
5. Write a program to sort an array using Insertion Sort.
6. Write a program to sort an array using Selection Sort.
7. Write a program to sort an array using Bubble Sort.
8. Write a program to search an element in an array using Binary Search.
9. Write a program to print the second largest and second smallest element in an array.
10. Write a program to reverse an array in-place.

Lab 9: Matrices & 2D Arrays

1. Write a program to add two matrices.
2. Write a program to multiply two matrices.
3. Write a program to find the sum of diagonal elements of a matrix.
4. Write a program to find the transpose of a matrix.
5. Write a program to print the upper and lower triangular parts of a matrix.
6. Write a program to check whether a matrix is symmetric or not.
7. Write a program to check whether a matrix is sparse or not.
8. Write a program to print the boundary elements of a matrix.
9. Write a program to find the row-wise and column-wise sum of a matrix.
10. Write a program to find the maximum element in each row and each column of a matrix.

Lab 10: Strings, Pointers, Files, Structures

1. Write a program to implement strlen() function.
2. Write a program to implement strcpy() function.
3. Write a program to implement strcat() function.
4. Write a program to count vowels and consonants in a string.
5. Write a program to check whether a string is palindrome or not.

6. Write a program to reverse a string without using library functions.
7. Write a program to count frequency of each character in a string.
8. Write a program to remove all spaces from a string.
9. Write a program to convert a string to uppercase and lowercase.
10. Write a program to search for a substring inside a string (manual strstr).
11. Write a program in C to count the words in a given file.
12. Write a program to perform encryption and decryption on a file.